1. What is inferential statistics?

Inferential statistics is the branch of statistics that uses sample data to make generalizations or predictions about a larger population. It helps estimate population parameters such as mean and proportion without studying the entire population. It also allows researchers to test hypotheses and draw conclusions. Techniques like confidence intervals and hypothesis testing are widely used. It is essential in decision-making where complete data collection is impractical.

Formula:
Sample Mean: $\bar{x} = \Sigma x / n$
Sample Proportion: $\hat{p} = x / n$

2. What is hypothesis testing and its components?

Hypothesis testing is a statistical method used to evaluate assumptions about a population using sample data. It involves two hypotheses: the null hypothesis (H_0) and the alternative hypothesis (H_1). Key components include significance level (α), test statistic, and p-value.

Formula:
$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$

3. Explain confidence interval and critical value.

A confidence interval is a range of values used to estimate a population parameter with a certain level of confidence.

Formula:
Confidence Interval (Z): $\bar{x} \pm Z(\alpha/2) * (\sigma / \sqrt{n})$
Confidence Interval (t): $\bar{x} \pm t(\alpha/2) * (s / \sqrt{n})$

4. Define p-value.

The p-value is the probability of obtaining results at least as extreme as the observed results, assuming the null hypothesis is true.

Formula:
p-value = $P(\text{observed result} \mid H_0 \text{ true})$

5. Differentiate Type I and Type II errors.

Type I error occurs when the null hypothesis is true but rejected. Type II error occurs when the null hypothesis is false but not rejected.

Formula:
Type I Error: α
Type II Error: β

6. Brief descriptions of tests.

Z-test, t-test, chi-square test, and ANOVA are used based on data type.

Formulas:
$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$
$t = (\bar{x} - \mu) / (s / \sqrt{n})$
$\chi^2 = \Sigma (O - E)^2 / E$
F = Variance Between / Variance Within

7. What is Covariance?

Covariance measures how two variables change together.

Formula:
$Cov(X,Y) = \Sigma (X - \bar{X})(Y - \bar{Y}) / n$

8. What is Correlation?

Correlation measures the strength and direction of relationship between two variables.

Formula:
$r = Cov(X,Y) / (\sigma X * \sigma Y)$